

Population = 1000
Sensors = 100
Islands = 5
Generation = 100
Threads per block = 512

Cuda kernel

Regular Function

Init both (population,
buffer) populations

Create shared memory
object

initBufferKernel (d_pop1)

$\text{Blocks} = (1000 + 512 - 1) / 512 = 2,951 = 3$

Threads/Block = 512

setupCurand

island size = $1000 / 5 = 200$

$\#block/island = (200 + 512 - 1) / 512 = 1,389 = 1$

Blocks = $5 * 1 = 5$

Create cudaStreams

Generations

initIslandBufferKernel

island size = $1000 / 5 = 200$

$\#block/island = (200 + 512 - 1) / 512 = 1,389 = 1$

Blocks = $5 * 1 = 5$

buildRouletteKernel

Blocks = 1

galslandKernel

island size = $1000 / 5 = 200$

$\#block/island = (200 + 512 - 1) / 512 = 1,389 = 1$

Blocks = $5 * 1 = 5$

